Full Stack Assignment – 22

Assignment ID: FSD-22

Title: Leave Management System

Duration: 3 Hours

1️⃣ Objective

Design and develop a **Leave Management System** that allows employees to apply for leave and managers to review and act on leave requests.

The system must enforce leave balance validation and ensure **data persistence using a database**.

2️⃣ User Roles

EMPLOYEE

- Register and log in to the system
- Apply for leave
- View own leave requests and their status

MANAGER

- Log in to the system
- View all leave requests
- Approve or reject leave requests

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Leave Management

- EMPLOYEE can apply for leave with date range and reason
- Leave requests must be stored in the database
- MANAGER can approve or reject leave requests
- Leave status must be retrieved dynamically

Leave Balance

- Each employee must have a leave balance
- Leave balance must be stored and updated in the database

4️⃣ Business Rule (Critical)

⚠️ **Leave requests must be rejected if the employee's leave balance is insufficient.**

Leave balance must be validated before approval.

5️⃣ Technical Constraints (Mandatory)

- All leave and balance data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for EMPLOYEE and MANAGER
- Leave application form for EMPLOYEE
- Leave request listing with status for MANAGER

UI design is not evaluated; data correctness and rule enforcement matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints

- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8️⃣ Evaluation Note

Evaluation focuses on **validation logic, role enforcement, and end-to-end integration**.